C^0-rigidity in contact topology via microlocal sheaves

joint work with Tomohiro Asano (Kyoto), Yuichi Ike (Tokyo) and Christopher Kuo (Max-Planck Bonn)

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C^0-symplectic and contact topology

Theorem (Gromov–Eliashberg, '85–'87)

Let (M, ω) be a symplectic manifold. Then $Symp(M, \omega) \subset Diff(M)$ is closed under the C^0-topology.

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Theorem (Laudenbach–Sikorav, '92, Leclercq–Humilière–Seyfaddini, '13)

Let $L_n \subset (M, \omega)$ be closed Lagrangians. Suppose $L_n \to L$ in the Hausdorff limit and $L \subset (M, \omega)$ is a half-dimensional smooth manifold. Then L is also Lagrangian.
Let $C \subset (M, \omega)$ be coisotropic. Suppose $\varphi_n \in Symp(M, \omega)$ and $\varphi_n \to \varphi$ in $Homeo(M)$ in the C^0-topology and $\varphi(C)$ is a smooth manifold. Then $\varphi(C)$ is also coisotropic.

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- However, it seems that in contact topology, there are more flexibility due to the *h*-principle (for Legendrian submanifolds, for instance).

Theorem (Fuchs–Tabachnikov, '95, Murphy, '12)

Let $\Lambda \subset (Y,\xi)$ be n-manifolds in $(2n+1)$-contact manifolds. Then there exists a sequence of Legendrians $\Lambda_n \to \Lambda$ in the Hausdorff limit.

C^0-symplectic and contact topology

- When the contact manifold has dimension at least 5, Murphy's result actually constructs a sequence of Legendrians $\Lambda_n \to \Lambda$ such that are Λ_n are all Legendrian isotopic (hence contactomorphic) to each other.

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- While this almost suggests that it may not be the case that C^0-limits of contactomorphisms send Legendrians to Legendrians, it was recently proved by Dimitroglou Rizell–Sullivan that this is the case for closed Legendrians, after works of Rosen–Zhang, Usher, Nakamura, and Stokic from '18 to '23.

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Theorem (Dimitroglou Rizell–Sullivan, '22)

Let $\Lambda \subset (Y, \xi)$ be closed Legendrians. Suppose $\varphi_n \in Cont_c(Y, \xi)$ are supported in one compact set and $\varphi_n \to \varphi$ in $Homeo(Y)$ in the C^0-topology and $\varphi(\Lambda)$ is a smooth manifold. Then $\varphi(\Lambda)$ is also Legendrian.

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Let $C \subset (Y, \xi)$. We say C is a coisotropic submanifold if $TC \cap \xi$ is coisotropic (but not necessarily constant rank).

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Theorem (Usher, '20)

Let $C \subset (Y, \xi)$ be coisotropics. Suppose $\varphi_n \in Cont(Y, \xi)$ and $\varphi_n \to \varphi$ in $Homeo(Y)$ in the C^0-topology, such that the conformal factors f_n (defined by $\varphi_n^* \alpha = e^{f_n} \alpha$) are uniformly bounded below, and suppose $\varphi(C)$ is a smooth manifold. Then $\varphi(C)$ is also Legendrian.

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Remark

A contactomorphism $\varphi : (Y, \xi) \to (Y, \xi)$ with conformal factor f lifts to $(Y \times \mathbb{R}, e^t \alpha) \to (Y \times \mathbb{R}, e^t \alpha)$, $(x, t) \mapsto (\varphi(x), t - f(\varphi(x)))$.

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Theorem (Abouzaid–Kragh '11–13, Guillermou '12, Membrez–Opshtein '17, Jin '20 (Abouzaid–Courte–Guillermou–Kragh '20))

Let $L \subset T^*M$ be a closed exact Lagrangian. Then L has trivial Maslov data (in fact, stronger results hold).
Let $\varphi : X \to X$ be a symplectic homeomorphism. Then the Maslov class of a closed Lagrangian L and $\varphi(L)$ agree.

C^0-symplectic and contact topology

Theorem (Dimitroglou Rizell–Sullivan, '22)

Let Y be a contact 3-manifold. Suppose $\varphi_n \in Cont_c(Y, \xi)$ are supported in one compact set and $\varphi_n \to \varphi$ in $Homeo(Y)$ in the C^0-topology. Then a closed Legendrian Λ and $\varphi(\Lambda)$ are related by an ambient contactomorphism.

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Theorem (Dimitroglou Rizell–Sullivan, '18 (Lazarev, '18-19))

Suppose $\varphi_n \in Cont_c(J^1M, \xi)$ are supported in one compact set and $\varphi_n \to \varphi$ in $Homeo(J^1M)$ in the C^0-topology. Let Λ be a closed Legendrian. Suppose Λ has an augmentation, then $\varphi(\Lambda)$ is not loose or stabilized.

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- We do not know whether C^0-limits of contactomorphisms send coisotropics to coisotropics in full generality.
- We do not know whether C^0-limits of contactomorphisms preserves Maslov data of Legendrians beyond dimension 3.
- We also do not know whether C^0-limits of contactomorphisms preserves Floer theory or SFT invariants of Legendrians.

Microlocal sheaf theory

- We propose an approach to understanding the C^0-rigidity problems in symplectic and contact topology in terms of microlocal sheaf theory, following Guillermou '13, Guillermou–Viterbo '22 and Asano–Ike '22.

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- Given a sheaf $F \in Sh(M)$, one can define a notion called **singular support** of the sheaf $SS^\infty(F)$, which is a closed subset in S^*M.
- Given a closed subset $\Lambda \subset S^*M$, we consider the full sub-category of sheaves $Sh_\Lambda(M)$ whose singular supports are contained in Λ.

Microlocal theory of sheaves

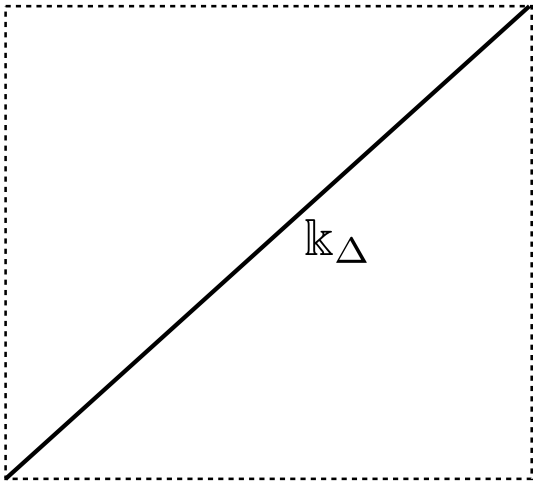
Example

(1) Consider a closed submanifold $N \subset M$, the constant sheaf k_N has singular support being the conormal bundle $S^*_N M$.
(2) Consider an open (resp. closed) tubular neighborhood of $U_r(N) \subset M$ (resp. $\overline{U}_r(N) \subset M$), the constant sheaf $k_{U_r(N)}[-n]$ (resp. $k_{\overline{U}_r(N)}$) has singular support being the outward (resp. inward) conormal of $U_r(N)$, i.e. the pushoff of $S^*_N M$ along the geodesic flow by r (resp. $-r$).

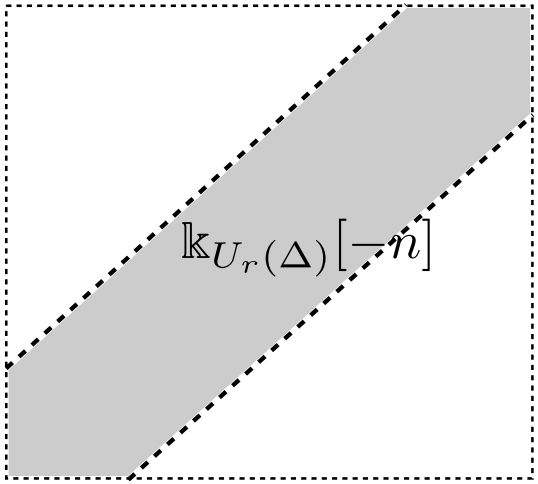
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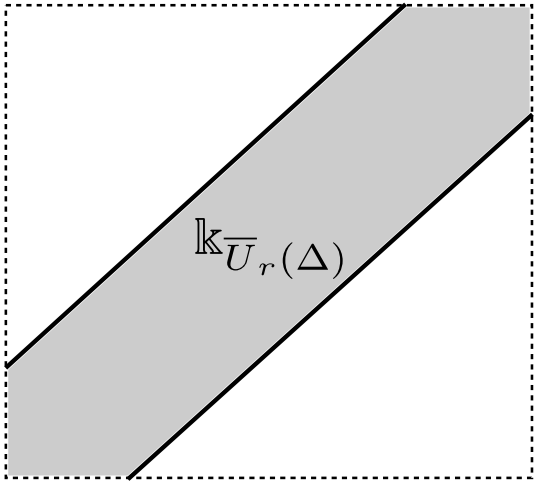
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Theorem (Guillermou '12, Jin '20)

Let $\Lambda \subset S^*M$ be a smooth Legendrian. The microlocalization, i.e. restriction of $Sh_\Lambda(M)$ along $\Lambda \subset S^*M$ defines a local system of categories twisted by the Maslov data of Λ.

Sheaf theory approach to C^0-rigidity

Theorem (Kashiwara–Schapira, '90, Guillermou–Viterbo, '22)

The singular support of any sheaf $F \in Sh(M)$ is singular coisotropic (γ-coisotropic or locally rigid).

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Given a contact isotopy φ_t, there is an equivalence $Sh_\Lambda(M) \simeq Sh_{\varphi_t(\Lambda)}(M)$ induced by a sheaf $K_\varphi \in Sh(M \times M)$ whose singular support is the graph of φ in $S^*M \times S^*M \times \mathbb{R} \subset S^*(M \times M)$.

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Theorem (Guillermou–Viterbo, Asano–Ike, '22)

There is a (right invariant) interleaving distance on $Sh(M \times M)$ (defined in terms of continuation maps of a Reeb flow). $Sh(M \times M)$ is complete with respect to the distance, i.e. any Cauchy sequence converges.

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- By completeness, we know K_{φ_n} converges to K_φ. By coisotropicity of singular supports, we know it sends Legendrians to coisotropics, if not empty (any Legendrian can locally be realized as the singular support, by Kashiwara–Schapira).

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- Suppose K_φ are invertible. Then $Sh_\Lambda(M)$ and $Sh_{\varphi(\Lambda)}(M)$ (hence the wrapped Fukaya categories) are equivalent. By microlocalization, the Maslov data are the same.

C^0-continuity of interleaving distance

Proposition

For compactly supported contactomorphisms φ on J^1M in the identity component with sheaf quantizations K_φ,

$$d(K_{id}, K_\varphi) \leq C d_{C^0}(id, \varphi).$$

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- For similar tricks, see for example Buhovsky–Humilière–Seyfaddini, Dimitroglou Rizell and Serraille–Stojsavljević.

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Theorem

Suppose M has bounded geometry (positive injectivity radius). There exists $\epsilon > 0$, so that for any contactomorphism φ on S^*M with $d_{C^0}(id, \varphi) < \epsilon$, there is a sheaf quantization K_φ with

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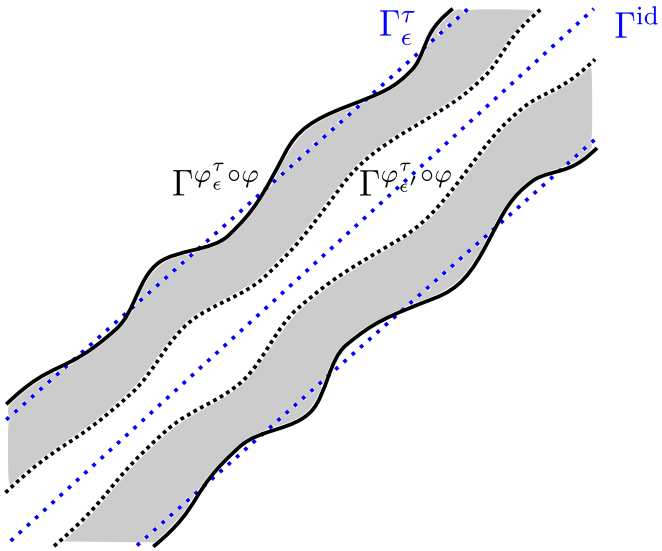
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- For a C^0-small contactomorphism φ, its graph is a Legendrian in the tubular neighborhood $U_r(\Delta_{S^*M})$ with no Reeb chords.

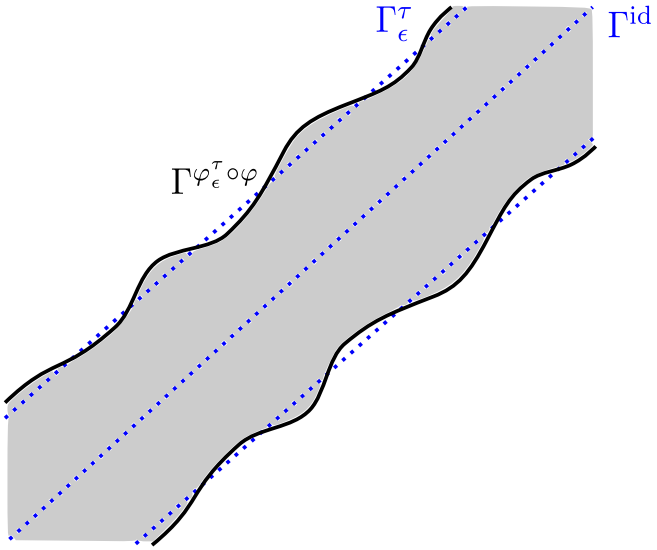
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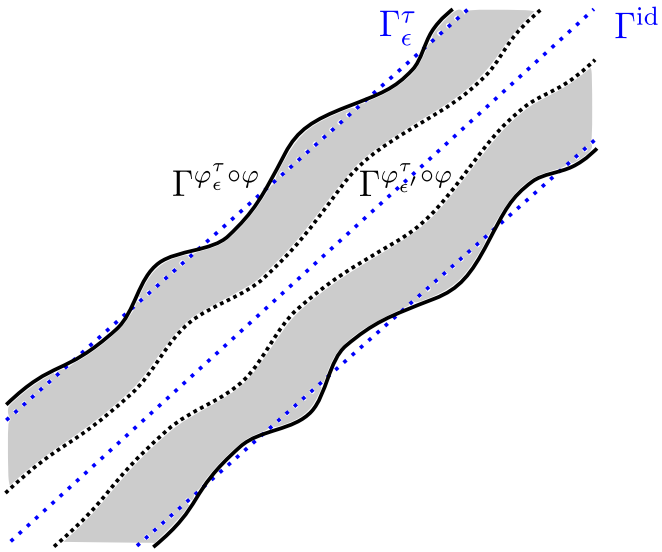
- For a C^0-small contactomorphism φ, its graph is a Legendrian in the tubular neighborhood $U_r(\Delta_{S^*M})$ with no Reeb chords.
- For a nearby Legendrian with no Reeb chords, Guillermou produces a sheaf quantization and showed that the interleaving distance from the zero section is bounded by the primitive/the r-coordinate.

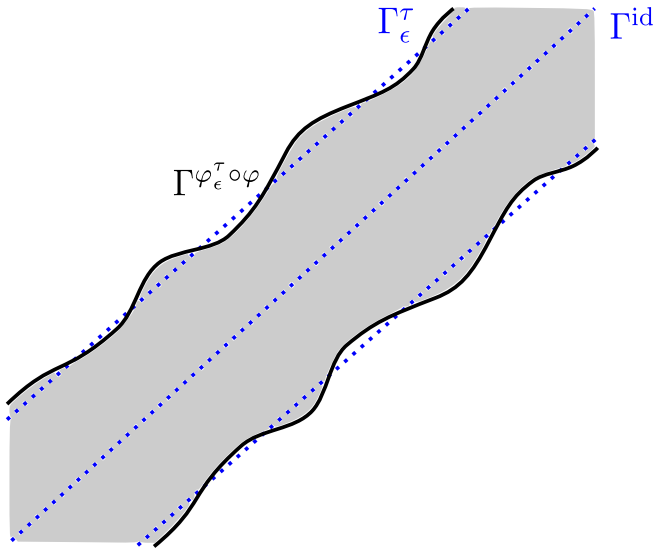




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- Sheaves associated to nearby Lagrangians in $U_r(\Delta_{S^*M})$ are isomorphic to the constant sheaf near boundary, so sheaf theoretically we can glue the Guillermou sheaf quantization in $U_r(\Delta_{S^*M})$ with the constant sheaf in $U_r(\Delta_M)$.





Showing the sheaf quantization is invertible

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- Finally, the distance of $K_{\varphi_n} \circ K_{\varphi^{-1}}$ and K_{id} is controlled by $d_{C^0}(id, \varphi_n \circ \varphi^{-1})$, so $K_{\varphi_n} \circ K_{\varphi^{-1}} \to K_{id}$.

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- Suppose $\Lambda \subset S^*M$ is closed Legendrian and $Sh_\Lambda(M)$ is *n*on-trivial. Suppose Λ_n are Legendrian isotopic to Λ such that they agree in a neighborhood of $p \in \Lambda$. Suppose $\Lambda_n \to \Lambda_\infty$ in the Hausdorff limit. Then Λ_∞ is Legendrian if smooth and has the same Maslov data.

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- Note that the non-trivial condition on $Sh_\Lambda(M)$ excludes loose Legendrians!
- The proof does not use spectral distance since we do not have global C^0-control. However, we can use the idea of Guillermou '13 to construct sheaves supported on Λ_∞:

$$F_\infty = \mathrm{Cone}\Big(\bigoplus_{n\in\mathbb{N}} F_n \to \prod_{n\in\mathbb{N}} F_n\Big).$$

Thank you!